Exercise 1: The interpreter

Open the Python interpeter. What happens when you input the following statements:

(a) 3 + 1

(b) 3 * 3

(c) 2 ** 3

(d) "Hello, world!"

Exercise 2: **Scripts**

Now copy the above to a script, and save it as script1.py. What happens if you run the script? (try: python script1.py). Can you fix this (hint: use the print function)

Exercise 3: **More interpreter**

Explain the output of the following statements if executed subsequently:

(a) 'py' + 'thon'

(b) 'py' * 3 + 'thon'

(c) 'py' - 'py'

(d) '3' + 3

(e) 3 * '3'

(f) a

(g) a = 3

(h) a

Exercise 4: **Booleans**

Explain the output of the following statements:

(a) 1 == 1

(b) 1 == True

(c) 0 == True

(d) 0 == False

(e) 3 == 1 * 3

(f) (3 == 1) * 3

(g) (3 == 3) * 4 + 3 == 1

(h) 3**5 >= 4**4

Exercise 5: Integers

Explain the output of the following statements:

(a) 5 / 3

(b) 5 % 3

(c) 5.0 / 3

(d) 5 / 3.0

(e) 5.2 % 3

(f) 2001 ** 200

Exercise 6: Floats

Explain the output of the following statements:

(a) 2000.3 ** 200 (compare with above)

(b) 1.0 + 1.0 - 1.0

(c) 1.0 + 1.0e20 - 1.0e20

Exercise 7: Variables

Write a script where the variable name holds a string with your name. Then, assuming for now your
name is John Doe, have the script output: Hello, John Doe! (and obviously, do not use print "Hello,
John Doe!".

Exercise 8: Type casting

Very often, one wants to \cast" variables of a certain type into another type. Suppose we have variable x = '123', but really we would like x to be an integer. This is easy to do in Python, just use desired type(x), e.g. int(x) to obtain an integer. Try the following and explain the output

(a) float(123)

(b) float('123')

(c) float('123.23')

(d) int(123.23)

(e) int('123.23')

(f) int(float('123.23'))

(g) str(12)

(h) str(12.2)

(i) bool('a')

(j) bool(0)

(k) bool(0.1)

Exercise 9: Hello

(a) Write a function hello_world that prints 'Hello, world!'

(b) Write a function hello_name(name) that prints 'Hello, name!' where name is a string.

(c) Explain the difference between the print and return keywords. What would change if instead of print you would use return?

Exercise 10: Polynomial

Write a function that evaluates the polynomial $3x^2 - x + 2$.